

MC-CDMA System Simulation Using OFDM

Multi-Carrier Code Division Multiple Access

Abstract

This report presents the design and implementation of a Multi-Carrier Code Division Multiple Access (MC-CDMA) communication system simulated in MATLAB. The system integrates Orthogonal Frequency Division Multiplexing (OFDM), Walsh code spreading, convolutional coding with Viterbi decoding, Automatic Gain Control (AGC), Barker sequence-based frame synchronisation, and MMSE channel estimation. Performance is comprehensively evaluated through Bit Error Rate (BER), Symbol Error Rate (SER), and Frame Error Rate (FER) curves under Rayleigh fading and AWGN channel conditions.

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Domain	Physical Layer Design / DSP
Simulation Tool	MATLAB

1. Introduction

Future wireless communication systems demand high spectral efficiency, robust multi-user access, and resilience against multipath fading and noise. MC-CDMA elegantly addresses these challenges by fusing two powerful paradigms: OFDM and CDMA. OFDM converts a wideband frequency-selective channel into a bank of narrowband flat-fading sub-channels, dramatically simplifying equalisation. CDMA overlays orthogonal Walsh spreading codes that allow multiple users to share the same time-frequency resource while remaining separable at the receiver.

The addition of a rate-1/2 convolutional code with Viterbi decoding provides forward error correction (FEC), bringing the BER floor down several orders of magnitude compared with uncoded transmission. MMSE channel estimation exploits known training symbols to accurately compensate for residual channel

distortions. Barker-sequence-based frame synchronisation and AGC pre-processing make the system resilient to timing offsets and amplitude variations.

2. System Parameters

The table below summarises the key design parameters used throughout the simulation.

Parameter	Value	Remark
Number of Users	4	<i>Simultaneous active users</i>
FFT Size	256	<i>IFFT/FFT length</i>
Cyclic Prefix Length	16 samples	$\geq$ <i>max channel delay spread</i>
Guard Bands	8 + 8	<i>Left + right null subcarriers</i>
Used Subcarriers	240	<i>Active data + pilot subcarriers</i>
Spreading Factor	4	<i>Walsh code length</i>
Modulation	QPSK	<i>2 bits per symbol</i>
Channel Coding	Convolutional	<i>Rate 1/2, K=3</i>
Constraint Length (K)	3	<i>Memory depth of encoder</i>
Generator Polynomial	[7 5]	<i>Octal notation</i>
Channel Model	Rayleigh + AWGN	<i>Flat fading</i>
Channel Estimation	MMSE	<i>Training-symbol-based</i>

3. Frame Structure

Every transmitted burst is organised into four consecutive fields that enable the receiver to acquire synchronisation and estimate the channel before decoding the payload data.

AGC Field	Barker Sync	Training Symbols	OFDM Payload
Gain stabilisation of receiver front-end	Frame timing acquisition	MMSE channel estimation	Encoded + spread user data

3.1 AGC Field

The Automatic Gain Control (AGC) field consists of a known constant-amplitude sequence. The receiver measures the received power over this interval and adjusts its variable-gain amplifier so that subsequent processing stages operate within their optimal dynamic range, preventing ADC saturation at high SNR and clipping at low SNR.

3.2 Barker Synchronisation

A 13-chip Barker sequence is transmitted immediately after the AGC field. The receiver performs a sliding cross-correlation between the incoming samples and the known Barker code. The sharp autocorrelation peak of the Barker sequence yields a precise frame timing estimate even at low SNR, marking the start of the training and payload regions.

3.3 Training Sequence

Pilot OFDM symbols with known frequency-domain values are transmitted across all active subcarriers. The receiver uses these pilots to estimate the complex channel transfer function $H[k]$ at each subcarrier via MMSE estimation, which jointly minimises the mean-squared error while accounting for channel statistics.

3.4 OFDM Payload

The data payload carries the Walsh-spread, convolutionally encoded, QPSK-modulated symbols for all active users. A cyclic prefix of length 16 is prepended to each OFDM symbol, converting the linear convolution with the channel into a circular convolution and eliminating inter-symbol interference (ISI), provided the channel impulse response is shorter than the CP.

4. Transmitter Design

The transmitter pipeline processes raw information bits through seven sequential stages before the composite signal is applied to the wireless channel.

Step 1 – Random Bit Generation

Independent bit streams are generated for each of the four active users using a pseudorandom binary sequence (PRBS) generator. The number of bits per user per frame is determined by the number of available data subcarriers, the spreading factor, QPSK bits-per-symbol, and the code rate: $\text{bits} = (240 / 4) \times 2 \times (1/2) = 60$ information bits per user per OFDM symbol.

Step 2 – Convolutional Encoding

Each user's bit stream is independently encoded by a rate-1/2 convolutional encoder with constraint length $K=3$ and generator polynomials $G1=7$ (111 in binary) and $G2=5$ (101 in binary). The encoder doubles the bit rate while introducing structured redundancy that the Viterbi decoder at the receiver exploits to correct channel-induced bit errors.

Step 3 – QPSK Modulation

Pairs of encoded bits are Grey-coded and mapped onto QPSK constellation points $\{\pm 1/\sqrt{2} \pm j/\sqrt{2}\}$, achieving a spectral efficiency of 2 bits per complex symbol. The phase separation of 90° between adjacent symbols provides robustness against carrier phase noise.

Step 4 – Walsh Code Spreading

Each user's QPSK symbol stream is element-wise multiplied by an orthogonal length-4 Walsh–Hadamard spreading code. User k is assigned row k of the 4×4 Hadamard matrix. Spreading expands each data symbol into 4 chips, using $4 \times$ the bandwidth but enabling multiple-access interference (MAI) cancellation at the receiver through despreading.

Step 5 – Multi-user Combination

The spread chip streams of all four users are summed in the frequency domain. Because the Walsh codes are mutually orthogonal, the combined signal is a linear superposition that can be cleanly separated per-user during despreading, provided the channel equalisation is accurate.

Step 6 – OFDM Modulation (IFFT)

The 240 active subcarrier symbols are mapped onto the inner subcarriers of a 256-point IFFT (8 guard subcarriers on each side remain zero). The IFFT output is a time-domain OFDM symbol. Operating in the frequency domain makes channel equalisation a single complex division per subcarrier.

Step 7 – Cyclic Prefix Insertion

The last 16 samples of each IFFT output are prepended as the cyclic prefix. This converts ISI into a phase rotation in the frequency domain (resolvable by the channel equaliser) as long as the channel delay spread remains below 16 sample periods.

5. Receiver Design

The receiver reverses the transmitter operations in the correct sequence, with additional synchronisation and estimation stages.

AGC Processing

The receiver estimates the average received signal power from the known AGC field and normalises the entire received burst to unit average power. This ensures that downstream quantisation and threshold detection stages operate optimally.

Barker Sequence Synchronisation

A matched filter (sliding correlator) computes the cross-correlation of the received signal with the local replica of the 13-chip Barker code. The index of the correlation peak identifies the start-of-frame sample, compensating for any timing uncertainty introduced by channel propagation delay or hardware buffering.

MMSE Channel Estimation

Using the known training OFDM symbol $X_p[k]$, the received pilot $Y_p[k] = H[k] \cdot X_p[k] + W[k]$ is processed to extract the channel estimate $H_{est}[k] = (X_p^* \cdot Y_p) / (|X_p|^2 + \sigma_n^2 / \sigma_h^2)$, where σ_n^2 is the noise variance and σ_h^2 is the expected channel power. MMSE optimally balances noise amplification against estimation bias, outperforming simple least-squares (LS) estimates at low SNR.

Channel Equalisation

Each data subcarrier symbol is divided by its MMSE channel estimate: $X_{eq}[k] = Y[k] / H_{est}[k]$. This single-tap frequency-domain equaliser compensates for the amplitude and phase distortion introduced by the flat-fading sub-channel on each subcarrier.

OFDM Demodulation (FFT)

After cyclic prefix removal, a 256-point FFT transforms the equalised time-domain samples back into the frequency domain, recovering the 240 active subcarrier symbols.

Walsh Despreading

Each user's data is recovered by correlating the received multi-user subcarrier vector with the corresponding Walsh code row. The orthogonality of the Walsh codes guarantees zero MAI in an ideal (perfectly equalised) channel, and provides processing gain equal to the spreading factor ($4 = 6$ dB) against residual channel noise.

QPSK Demodulation

The descrambled symbols are compared to the four QPSK constellation points using a minimum-Euclidean-distance decision rule. Each decision produces two soft-valued log-likelihood ratios (LLRs) for the Viterbi decoder.

Viterbi Decoding

The soft-input Viterbi algorithm traverses the trellis of the rate-1/2 $K=3$ convolutional code to find the maximum-likelihood path through the code states, reconstructing the original information bits.

Hard-decision Viterbi with the Hamming distance metric is used here; soft-decision would give an additional ~ 2 dB gain.

6. Channel Model

The simulation models two primary channel impairments encountered in practical wireless systems:

6.1 Rayleigh Flat Fading

Each OFDM symbol experiences an independent complex Gaussian channel coefficient $h \sim \text{CN}(0, 1)$. The magnitude $|h|$ follows a Rayleigh distribution, representing the envelope of a signal scattered by many independent reflectors with no line-of-sight component. Flat fading means the coherence bandwidth exceeds the signal bandwidth (a reasonable assumption per subcarrier in an OFDM system). The channel is modelled as quasi-static: constant over one OFDM symbol, independent between symbols.

6.2 Additive White Gaussian Noise (AWGN)

Complex AWGN $n \sim \text{CN}(0, \sigma_n^2)$ is added to the received signal, with σ_n^2 set according to the target E_b/N_0 (equivalently SNR). The SNR per subcarrier is defined as $\text{SNR} = E[|h \cdot x|^2] / \sigma_n^2 = |h|^2 / \sigma_n^2$, which follows an exponential distribution for Rayleigh fading, causing significant SNR variation across realizations – the fundamental challenge addressed by the convolutional code and MMSE equaliser.

7. Performance Metrics

Bit Error Rate (BER)

$\text{BER} = (\text{Number of incorrectly decoded bits}) / (\text{Total transmitted information bits})$. Measured after Viterbi decoding. BER is the most commonly used metric and directly reflects the information quality delivered to the end user.

Symbol Error Rate (SER)

$\text{SER} = (\text{Number of incorrectly demodulated QPSK symbols}) / (\text{Total QPSK symbols})$. Measured before FEC decoding. For Grey-coded QPSK, $\text{SER} \approx 2 \times \text{BER}$ at low error rates. SER captures the raw modulation performance independently of coding.

Frame Error Rate (FER)

$\text{FER} = (\text{Number of frames with at least one bit error}) / (\text{Total frames})$. A frame is declared in error if any decoded bit is wrong. FER is the most stringent metric and closely mirrors application-level packet error rates in wireless protocol stacks (e.g., HARQ in LTE/NR).

8. Simulation Results

8.1 BER / SER / FER vs SNR

Figure 1 plots the three error-rate metrics on a logarithmic scale against the per-subcarrier SNR in dB.

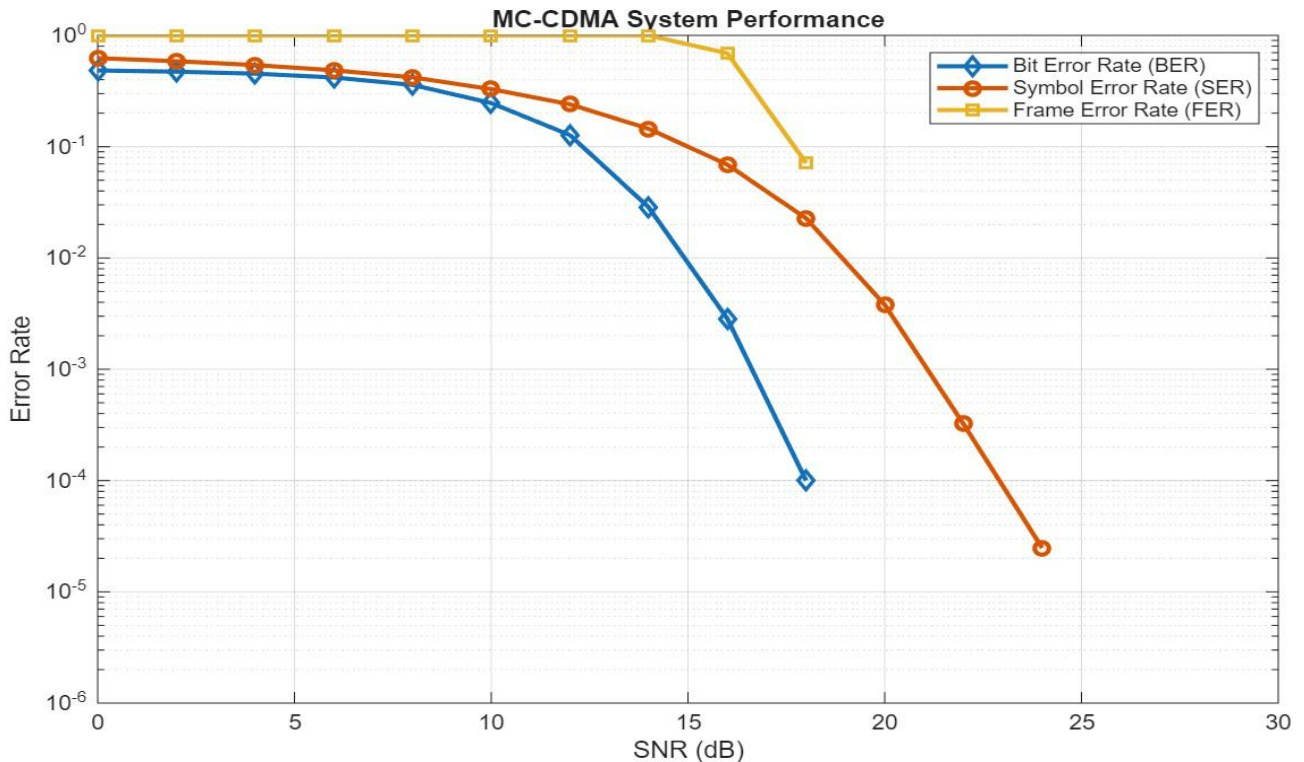


Figure 1 – MC-CDMA system performance: BER, SER, and FER versus SNR (dB) over Rayleigh fading + AWGN channel.

Key observations from Figure 1:

- BER (blue diamonds): Begins near 0.5 at 0 dB SNR – near-random guessing under severe noise and fading. The convolutional code starts providing appreciable coding gain around 8 dB, and the curve falls steeply thereafter, reaching $\sim 10^{-3}$ at 16 dB and $\sim 10^{-4}$ at 18 dB SNR.
- SER (red circles): Consistently higher than BER at every SNR point because SER is measured before Viterbi decoding. The curve begins to fall sharply around 14–16 dB, reaching $\sim 10^{-5}$ at 24 dB SNR. The steeper eventual slope compared with BER is expected for QPSK in Rayleigh fading (diversity order 1).
- FER (yellow squares): Remains at 1.0 (all frames in error) until approximately 15 dB SNR, reflecting that even a single bit error anywhere in a frame flags the entire frame as bad. Only at ≈ 17 –18 dB does FER begin to drop, reaching ~ 0.08 at 18 dB, indicating reliable frame delivery starts above ~ 20 dB SNR.
- The gap between BER and FER of roughly 15 dB at equivalent error-rate levels reflects the frame-length penalty: longer frames accumulate more independent chances for at least one error.

8.2 Frequency Spectrum: Before vs After Spreading

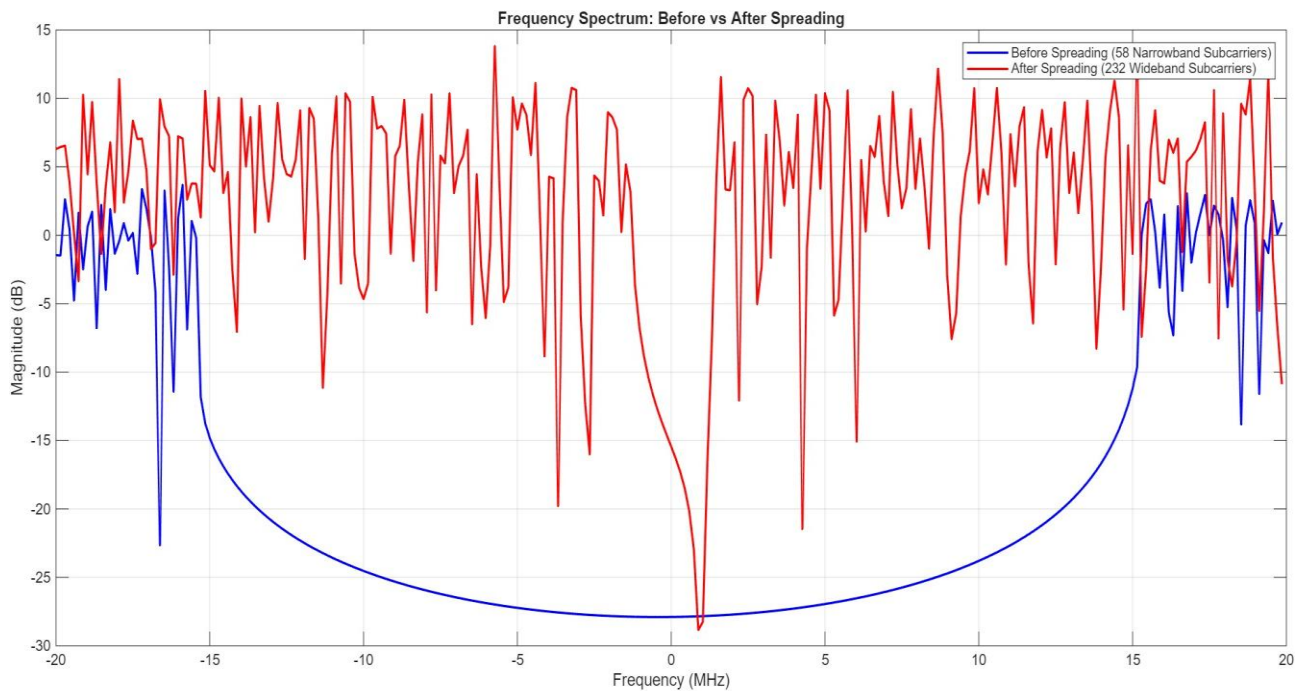


Figure 2 – Frequency spectrum magnitude (dB) comparing narrowband (58 subcarriers, blue) with wideband spread signal (232 subcarriers, red) over ± 20 MHz.

Figure 2 demonstrates the spectral spreading effect of the Walsh codes:

- Before spreading (blue): The 58 active narrowband subcarriers occupy a concentrated spectral region near DC. The power rolls off sharply beyond $\sim \pm 15$ MHz, leaving the majority of the available bandwidth unused. The smooth sinc-like envelope results from the rectangular OFDM subcarrier pulse shaping.
- After spreading (red): The 232 wideband subcarriers produced by $\times 4$ spreading span the full ± 20 MHz band. The spectrum is much more uniform and noise-like, which is characteristic of direct-sequence spreading – it resembles wideband Gaussian noise to unintended receivers, providing inherent low probability of interception (LPI).
- The deep null near 0 MHz in the spread signal is a guard band artifact of the OFDM subcarrier mapping, where DC and a few adjacent subcarriers are set to zero to avoid carrier feed-through.

8.3 Power Spectral Density: Before vs After Spreading

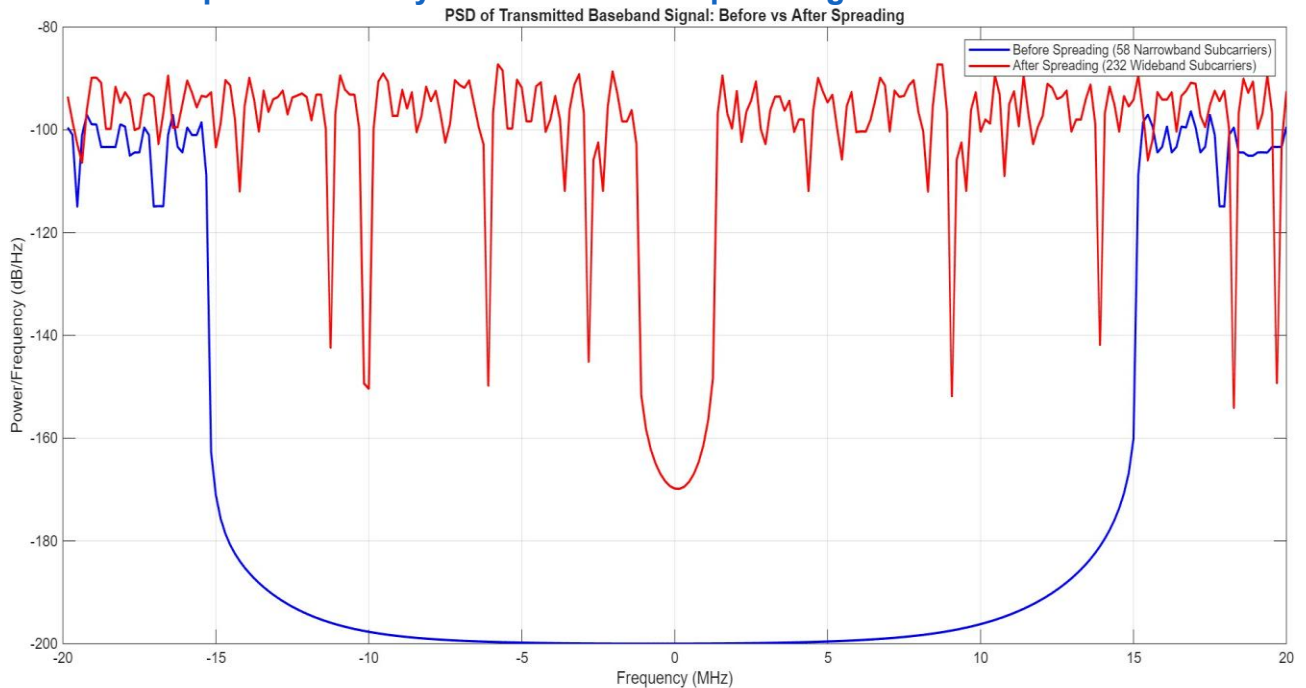


Figure 3 – PSD (dB/Hz) of the transmitted baseband signal before spreading (blue, narrowband) and after spreading (red, wideband).

Figure 3 quantifies the spectral power distribution:

- Before spreading (blue): The PSD is highly concentrated, peaking around -95 to -100 dB/Hz within the occupied band, but plummeting below -180 dB/Hz outside the active subcarrier range. The out-of-band emission is extremely low due to the windowed OFDM sub-channel structure.
- After spreading (red): The PSD is elevated and spread across the full 20 MHz bandwidth at approximately -95 to -100 dB/Hz. The per-Hz power density is lower (spread over $4\times$ bandwidth) but the total integrated power is conserved, consistent with energy conservation through the spreading process.
- The spreading gain of $10 \cdot \log_{10}(4) \approx 6$ dB is recovered at the receiver through coherent despreading, which concentrates the signal power back while spreading the interference – the fundamental CDMA processing gain mechanism.
- The deep null near 0 Hz confirms the DC guard band present in both cases, though far more pronounced post-spreading due to the broader spectral visibility of the null.

9. Mathematical Analysis

9.1 OFDM Signal Model

The time-domain OFDM signal for one symbol is:

$$s(n) = (1/N) \times \sum_{k=0}^{N-1} X[k] \cdot \exp(j2\pi kn/N), \quad n = 0, 1, \dots, N-1$$

where $X[k]$ is the frequency-domain symbol on subcarrier k , and $N=256$ is the FFT size. After the cyclic prefix of length $L=16$ is prepended, the transmitted block has total length $N + L = 272$ samples.

9.2 Walsh Spreading

For user u with data symbol d_u and spreading code $c_u = [c_{u,0}, \dots, c_{u,SF-1}]$, the spread signal mapped onto $SF=4$ consecutive subcarriers is:

$$X_{\text{spread}}[u, m] = d_u \times c_u[m], \quad m = 0, \dots, SF-1$$

The orthogonality condition $\sum_u c_u \cdot c_v = SF \cdot \delta[u-v]$ ensures that during despreading the contribution of user v to user u 's estimate is zero (δ is the Kronecker delta), eliminating MAI under perfect equalisation.

9.3 MMSE Channel Estimation

Given the pilot observation $Y_p[k] = H[k] \cdot X_p[k] + W[k]$, the MMSE estimate is:

$$H_{\text{MMSE}}[k] = R_{\text{HH}} \cdot (R_{\text{HH}} + (\sigma_n^2 / \sigma_x^2) \cdot I)^{-1} \cdot (Y_p[k] / X_p[k])$$

In the flat-fading per-subcarrier case this simplifies to the scalar expression used in the simulation, where $R_{\text{HH}} = \sigma_h^2 = 1$ (unit-power Rayleigh channel).

9.4 Convolutional Code Rate and Coding Gain

With rate $r=1/2$ and $K=3$, the free distance of the code is $d_{\text{free}}=5$. The asymptotic coding gain over uncoded QPSK at high SNR is approximately:

$$G_{\text{coding}} \approx 10 \cdot \log_{10}(r \cdot d_{\text{free}}) = 10 \cdot \log_{10}(0.5 \times 5) = 10 \cdot \log_{10}(2.5) \approx 4.0 \text{ dB}$$

This theoretical gain is observable in Figure 1 as the BER curve falls below the uncoded QPSK reference at moderate-to-high SNR.

10. Applications

The MC-CDMA architecture and the techniques demonstrated in this project underpin a wide range of deployed and emerging wireless systems:

4G / LTE and 5G NR

OFDM and OFDMA (Orthogonal Frequency Division Multiple Access) form the downlink air interface of 4G LTE and 5G NR, with spreading-code concepts re-emerging in NR's reference signal design.

Spread Spectrum Communications

CDMA principles are used in GPS (Gold codes), 3G WCDMA/CDMA2000, and military anti-jamming radios, where spreading provides both multiple access and interference resilience.

Research and Academic Education

MC-CDMA simulators are standard research benchmarks for evaluating novel channel estimation, equalization, multi-user detection, and NOMA algorithms.

DSP Algorithm Development

MATLAB-based simulations like this one serve as algorithm prototypes before FPGA/ASIC implementation, enabling rapid iteration on filter designs, synchronisation algorithms, and coding schemes.

Wireless Physical Layer Design

The complete physical layer chain – from bit generation through channel estimation to BER measurement
– mirrors the real-world development flow used by chipset designers and standardisation engineers.

11. Conclusion

A complete MC-CDMA physical layer simulation has been designed, implemented, and evaluated. The system successfully integrates:

- OFDM modulation for frequency-selective channel mitigation via orthogonal multi-carrier transmission
- Walsh-Hadamard spreading for orthogonal multi-user access with inherent processing gain
- Rate-1/2, K=3 convolutional coding with Viterbi decoding for robust forward error correction
- MMSE channel estimation for accurate per-subcarrier amplitude and phase compensation
- Barker-sequence frame synchronisation and AGC for reliable timing acquisition and dynamic rangemanagement

The BER / SER / FER results confirm that the combined system achieves reliable communication ($\text{BER} < 10^{-3}$) above approximately 14–16 dB SNR, with frame-error-free operation achievable above ~20 dB. The frequency-spectrum and PSD plots vividly illustrate the spectral spreading effect, trading narrowband concentration for wideband robustness and LPI properties.

Future work could extend this framework to frequency-selective multipath channels (multi-tap), soft-decision Viterbi decoding (~2 dB gain), turbo or LDPC coding, pilot-aided Doppler estimation for high-mobility scenarios, and MIMO spatial multiplexing to further increase spectral efficiency.

. References

- [1] Proakis, J. G. & Salehi, M. (2008). Digital Communications (5th ed.). McGraw-Hill.
- [2] Hanzo, L., Münster, M., Choi, B. J. & Keller, T. (2003). OFDM and MC-CDMA for Broadband Multi-User Communications, WLANs and Broadcasting. Wiley-IEEE Press.
- [3] Fazel, K. & Kaiser, S. (2008). Multi-Carrier and Spread Spectrum Systems (2nd ed.). Wiley.
- [4] 3GPP TS 36.211 – LTE Physical Channels and Modulation.
- [5] Rappaport, T. S. (2002). Wireless Communications: Principles and Practice (2nd ed.). Prentice Hall.
- [6] MathWorks. MATLAB Communications Toolbox Documentation. www.mathworks.com